Predicting Premier League match results

Project phase 1: the idea, the data, the analyses we intend

Can public data forecast a match as well as the odds?

The task

Before kickoff, give every match three probabilities: home win, draw, away win.

The data

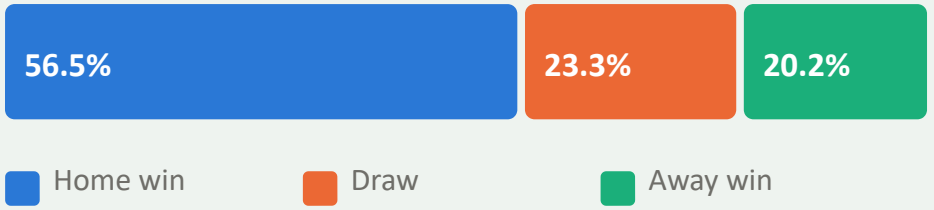
Twelve seasons of results, match statistics and bookmaker odds, all public.

The benchmark

The bookmaker's odds are a forecast too. We measure how close our models get.

What a forecast looks like

Arsenal v Chelsea



Expected goals **1.77 to 0.96**

Most likely score **1-0, with 11.6%**

Output of the first prototype, a Poisson goal model.

One CSV per season, fetched by a small crawler

Football-Data.co.uk

One CSV file per league and season

extraction/ download.py

One request per season

data/raw

12 files, 1.8 MB, saved untouched

data/final

Clean tables built by preprocess.py and features.py

How the crawler behaves

- Builds each address from the season: …/mmz4281/2324/E0.csv is the Premier League in 2023/24
- Names itself in the request and waits one second between requests
- Checks that the reply is a match file before saving it
- Skips files already downloaded, so running it again is safe

12 season files, 2014/15 to 2025/26

380 rows per file, one per match

62–132 columns per file, depending on the season

Cleaning turns twelve raw files into two tables

matches.csv

4,560 rows, 31 columns: one row per match

- Two date formats (dd/mm/yy and dd/mm/yyyy) read as one
- Blank and unplayed rows dropped
- Up to 132 raw columns reduced to 31, with readable names
- Result (H, D, A) derived from the score
- Columns a season lacks are left empty, never invented

features.csv

21 features per match, all known before kickoff

- Elo rating of each side, updated after every match
- Form over the last 5 matches: points, goals, shots on target
- Strength over the last 38 matches: goals scored and conceded
- Form at this venue, and days of rest
- Each feature uses earlier matches only, and a test checks it

The dataset at a glance

4,560
matches

12
seasons, 2014/15 to 2025/26

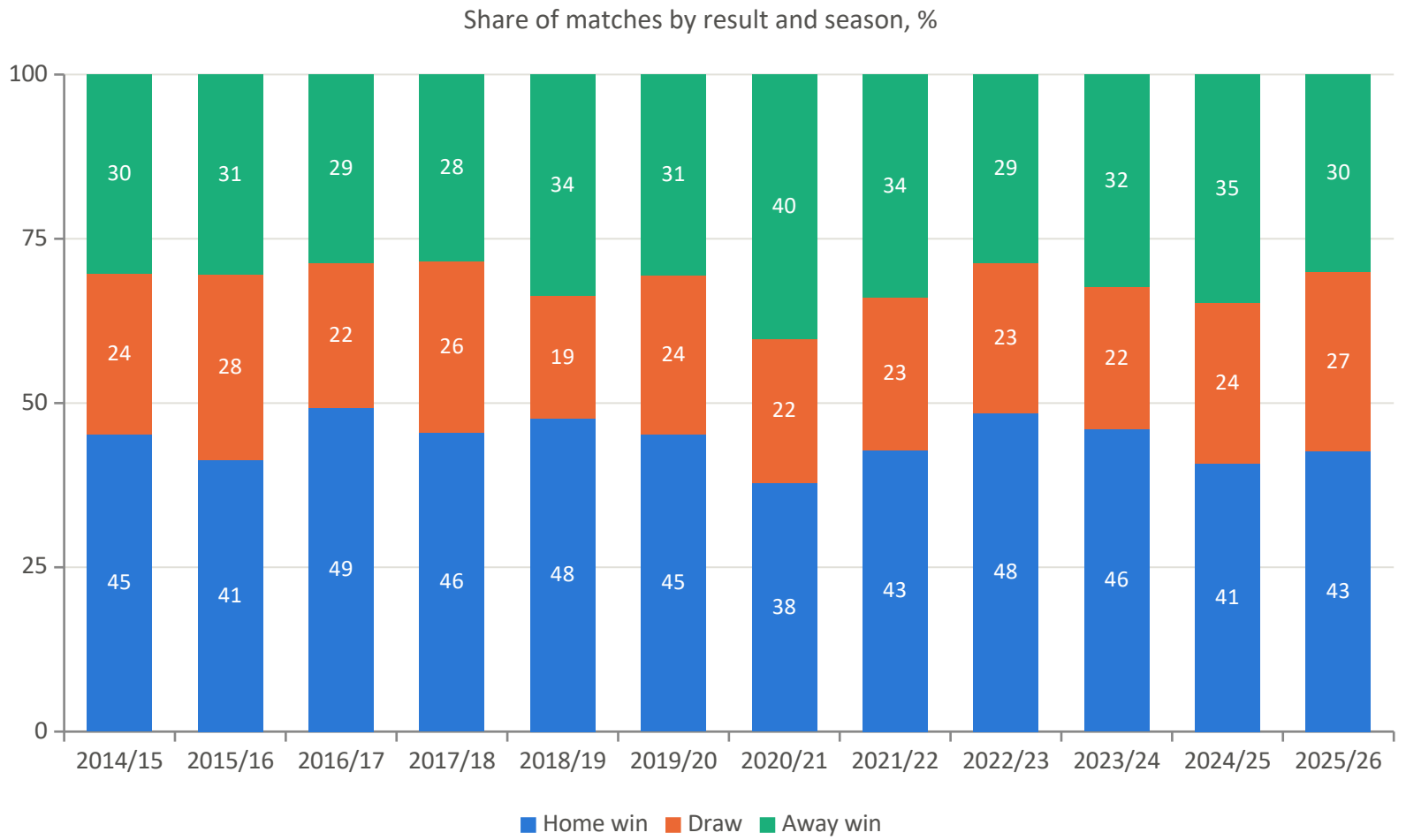
35
clubs

2.80
goals per match

44%
of matches won by the home side

Column group	Columns	Examples	Missing values
Identity	6	season, date, home, away, referee	none
Result	5	full-time and half-time goals, result	none
Match statistics	12	shots, shots on target, corners, fouls, cards	none
Bet365 odds, days before	3	home, draw, away	none
Bet365 odds at kickoff	3	home, draw, away	1,900 matches: published from 2019/20
Bet365 over/under 2.5 goals	2	over, under	1,901 matches

The home side wins 44% of matches



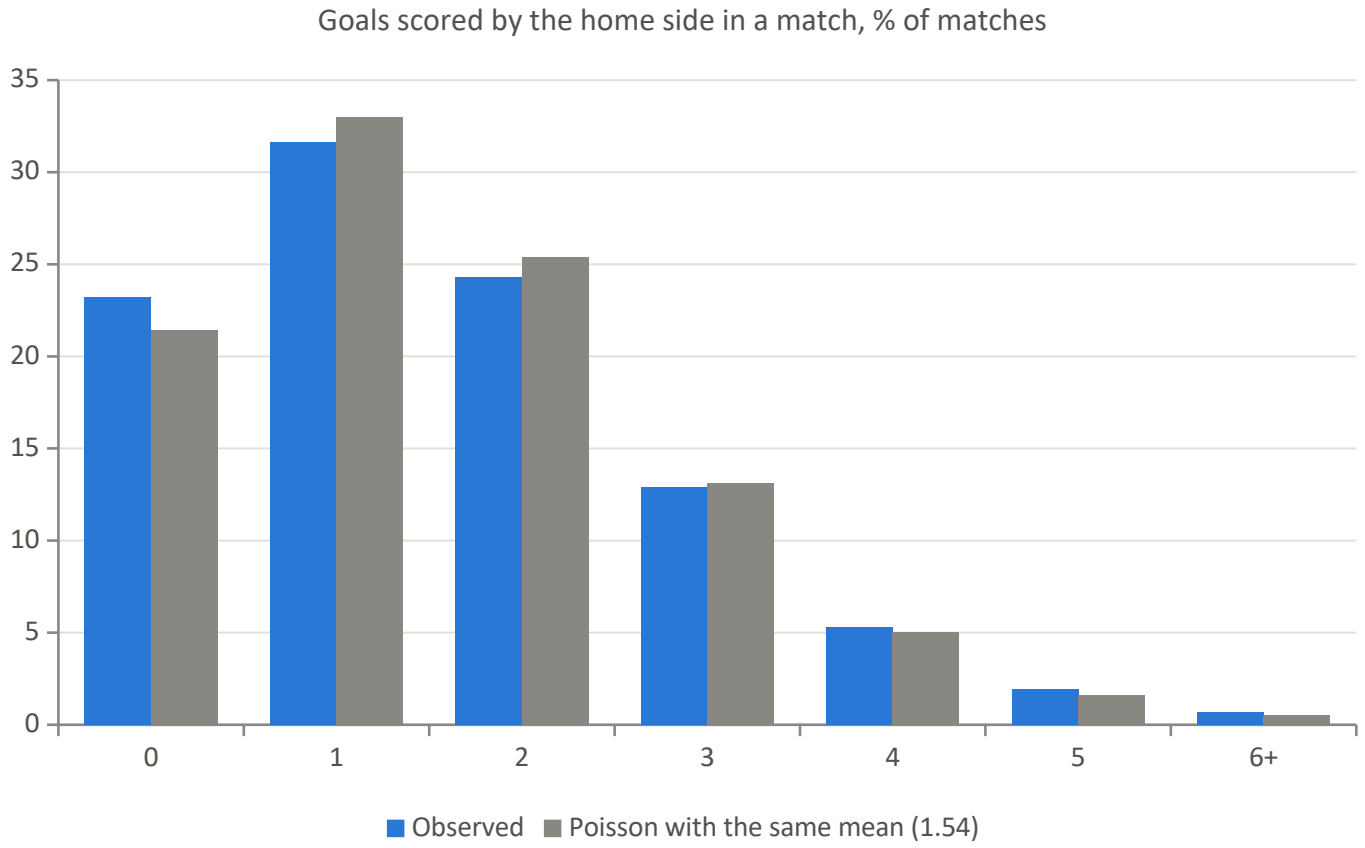
38%

home wins in 2020/21, played mostly without crowds. Away wins rose to 40%.

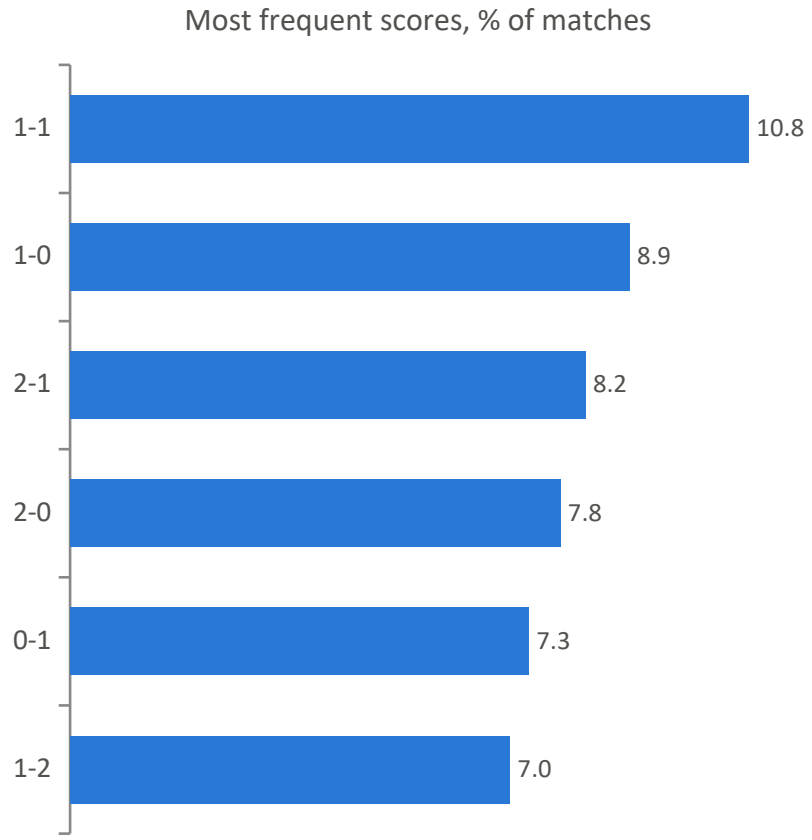
19–28%

of matches end in a draw, depending on the season.

Goals follow a Poisson distribution closely



Home sides average 1.54 goals, away sides 1.26. The variance (1.70 and 1.42) is only slightly above the mean.



54% of matches have more than 2.5 goals.

What we will do in the final phase

1 Explore

Home advantage over time, goal distributions, and which pre-match features relate to the result.

2 Model

Class frequencies as a floor, an Elo rating, a Poisson goal model, logistic regression, random forest and gradient boosting.

3 Evaluate

Walk-forward on the last three seasons, 1,140 matches. Accuracy, Brier score and log loss, against Bet365's odds at kickoff.

4 Analyse errors

Confusion matrix, calibration, and errors by kind of match and by club.

Repository

/extraction

crawling and preprocessing

/data

raw and final datasets

/docs

presentations and report

/analysis

models, in the final phase